

Note 1.3 Trigonometric Functions

The Six Basic Trigonometric Functions

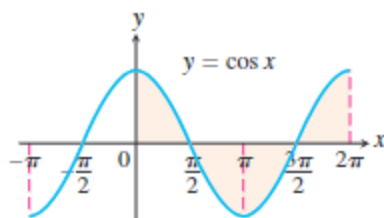
- Suppose in a triangle whose sides are adjacent, opposite and hypotenuse, then

- $\sin \theta = \frac{\text{opp}}{\text{hyp}}$
- $\cos \theta = \frac{\text{adj}}{\text{hyp}}$
- $\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin \theta}{\cos \theta}$
- $\csc \theta = \frac{\text{hyp}}{\text{opp}} = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{\text{adj}}{\text{opp}} = \frac{\cos \theta}{\sin \theta}$

- Suppose a circle, then

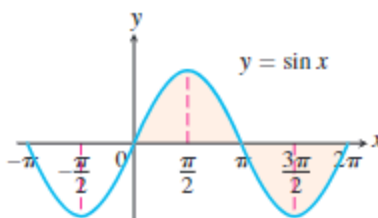
- $\sin \theta = \frac{y}{r}$
- $\cos \theta = \frac{x}{r}$
- $\tan \theta = \frac{y}{x} (x \neq 0)$
- $\csc \theta = \frac{r}{y}$
- $\sec \theta = \frac{r}{x}$
- $\cot \theta = \frac{x}{y} (y \neq 0)$

- Graphs:**



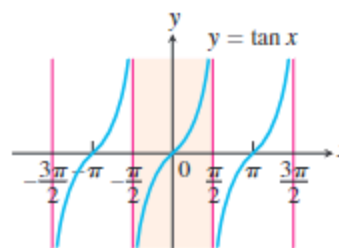
Domain: $-\infty < x < \infty$
Range: $-1 \leq y \leq 1$
Period: 2π

(a)



Domain: $-\infty < x < \infty$
Range: $-1 \leq y \leq 1$
Period: 2π

(b)

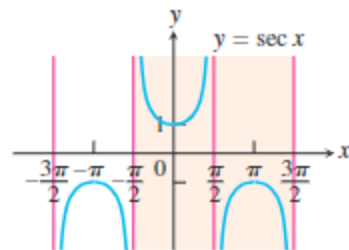


Domain: $x \neq \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$

Range: $-\infty < y < \infty$

Period: π

(c)

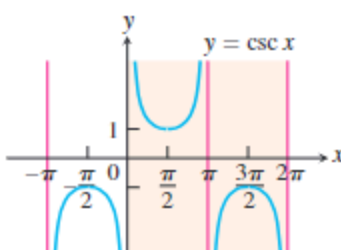


Domain: $x \neq \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$

Range: $y \leq -1$ or $y \geq 1$

Period: 2π

(d)

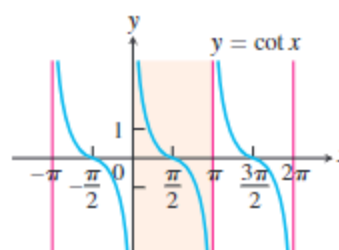


Domain: $x \neq 0, \pm \pi, \pm 2\pi, \dots$

Range: $y \leq -1$ or $y \geq 1$

Period: 2π

(e)



Domain: $x \neq 0, \pm \pi, \pm 2\pi, \dots$

Range: $-\infty < y < \infty$

Period: π

(f)

- **Period Function:**

A function $f(x)$ is periodic if there is a positive number p such that $f(x + p) = f(x)$ for every value of x . The smallest such value of p is the period of f .

Trigonometric Identities

- Pythagorean Theorem
- $\cos^2 \theta + \sin^2 \theta = 1, 1 + \tan^2 \theta = \sec^2 \theta, 1 + \cot^2 \theta = \csc^2 \theta$
- Addition Formulas
- Double-Angle Formulas
- Half-Angle Formulas
- The Law of cosines

Two special Inequalities

- $-|\theta| \leq \sin \theta \leq |\theta|$
- $-|\theta| \leq 1 - \cos \theta \leq |\theta|$